MATH 189 - Data Analysis

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Contents

1 Statistical Learning

1.1 Terminology

"Supervised Leanring" means we have both predictors and responses.
"Predictors/input variables/features" are arranged in a vector.

$$X = (X_1, X_2, ..., X_n)$$

"Output variables/responses" is a scalar where $y \in \mathbb{R}$
We think of data as coming from a model.

$$y = f(x) + \epsilon$$

$$\begin{cases} f(x) \text{ is fixed but unknown function} \\ \epsilon \text{ is the random error term with mean zero, independent of X} \end{cases}$$

Question: Why estimate f?
Prediction: Given X, we want to predict y with $\hat{y} = \hat{f}(x)$

Mean square error is

$$\begin{aligned}\mathbb{E}(y-\hat{y})^2 &= \mathbb{E}(f(x)+\epsilon-\hat{f}(x)) \\ &= \mathbb{E}(f(x)-\hat{f}(x))^2 + 2\mathbb{E}((f(x)-\hat{f}(x))\epsilon) + \mathbb{E}(\epsilon^2)\end{aligned}$$

Examine the middle term in detail

$$2\mathbb{E}((f(x)-\hat{f}(x))\epsilon) = \mathbb{E}(f(x)-\hat{f}(x)) \cdot \mathbb{E}(\epsilon) = 0$$

We know that

$$\mathbb{E}(y-\hat{y})^2 = (f(x)-\hat{f}(x))^2 + \mathbb{E}(\epsilon^2)$$

Where the left term is reducible error and the right term is irreducible error.

"Inference" - Sometimes we want to understand the association between y and $X_1, ..., X_p$

1. Which predictors are associated with response?
2. Is the association positive or negative?
3. Is the relationship approximately linear?

1.2 Accuracy

Training data

$$\begin{gathered}(x_1, y_1), (x_2, y_2), ...(x_n, y_n) \\ \text{where } X_i = (X_{i1}, X_{i2}, ..., X_{ip})\end{gathered}$$

n = # data points (e.g. $n = 30$)
p = # predictors (e.g. $p = 3$)

Mean Square Error (MSE)

$$\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{f}(x_i))^2$$

Defined on training data

$$\text{Test MSE} = \text{ Ave } (y_0 - \hat{f}(x_0))^2$$

Defined on previously unseen data point (x_0, y_0)

Linear / affine model

$$f(x) = \beta_0 + \beta_1 x_1 + ... + \beta_p x_p$$

"Fit/train the model" usually means run an optimization on your computer that uses training data and outputs the parameters $\beta_0, \beta_1, ..., \beta_p$ that make the MSE and hopefully the test MSE small.

Classification
Data set $(x_1, y_1), (x_2, y_2,), ..., (x_n, y_n)$ where $X_i = (x_{i1}, ..., x_{ip})$ and y_i is qualitative.

Error rate is the fraction of incorrect classifications.

$$\frac{1}{n}\sum_{i=1}^{n}\mathbb{I}\{y_i \neq \hat{y}_i\}$$

$$\mathbb{I}\{y_i \neq \hat{y}_i\} = \begin{cases} 1, & y_i \neq \hat{y}_i \\ 0, & y_i = \hat{y}_i \end{cases}$$

Test error rate

$$\text{Ave}(\mathbb{I}\{y_0 \neq \hat{y}_0\})$$

Where (x_0, y_0) is a previously unseen test data point.

K-nearest neighbors (KNN)
The KNN classifier is defined by letting η_0 = set of indices $i \in \{1, ..., n\}$ for K nearest neighbors of X_0 among the input data.

$\hat{y}_0 = j$ where j maximizes

$$\sum_{i \in \eta_0} \mathbb{I}\{y_i = j\}$$

which is the number of nearest neighbors (out of K) assigned label j.

2 Linear Regression

2.1 Simple Linear Regression

Underlying statistical model: $y = \beta_0 + \beta_1 X + \epsilon$ where y is the quantitative response and a single predictor X which is why it is called simple.

Prediction model: $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X$
$\hat{\beta}_0, \hat{\beta}_1$ are estimates based on training data, where hat means estimated quantity.

<u>Estimating coefficients</u>
The residual sum of squares (RSS)

$$\begin{aligned} \text{RSS} &= \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \\ &= \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \end{aligned}$$

Choose $\hat{\beta}_0, \hat{\beta}_1$ to minimize RSS.

$$\begin{aligned} \frac{\partial \text{RSS}}{\partial \hat{\beta}_0} &= \frac{\partial}{\partial \hat{\beta}_0} \sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 X)^2 \\ &= 2 \sum_{i=1}^{n} (\hat{\beta}_0 + \hat{\beta}_1 X_i - Y_i) \\ &\implies \text{RSS is } \begin{cases} \text{decreasing for really small } \hat{\beta}_0 \\ \text{increasing for really big } \hat{\beta}_0 \end{cases} \end{aligned}$$

RSS is optimal (in $\hat{\beta}_0$) when

$$
\begin{aligned}
0 &= \frac{1}{n}\sum_{i=1}^{n}(\hat{\beta}_0 + \hat{\beta}_1 X_i - Y_i) \\
&= \hat{\beta}_0 + \hat{\beta}_1\Big(\frac{1}{n}\sum_{i=1}^{n} X_i\Big) - \Big(\frac{1}{n}\sum_{i=1}^{n} Y_i\Big) \\
&= \hat{\beta}_0 + \hat{\beta}\bar{x} - \bar{y} \\
\hat{\beta}_0 &= \bar{y} - \hat{\beta}_1\bar{x}
\end{aligned}
$$

Where $\begin{cases} \bar{x} = \frac{1}{n}\sum_{i=1}^{n} X_i \text{ is the mean predictor} \\ \bar{y} = \frac{1}{n}\sum_{i=1}^{n} Y_i \text{ is the mean response} \end{cases}$

Covariance

$$
\begin{aligned}
\text{Cov}(X_i, Y_i)_{i=1}^{n} &= \frac{1}{n}\sum_{i=1}^{n} X_i Y_i - \bar{X}\bar{Y} \\
&= \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{x})(Y_i - \bar{y})
\end{aligned}
$$

Variance

$$
\begin{aligned}
\text{Var}(X_i)_{i=1}^{n} &= \text{Cov}(X_i, X_i)_{i=1}^{n} \\
&= \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{x})^2
\end{aligned}
$$

From this we can get our optimal $\hat{\beta}_1$

$$
\begin{aligned}
\hat{\beta}_1 &= \frac{\text{Cov}(X_i, Y_i)_{i=1}^{n}}{\text{Var}(X_i)_{i=1}^{n}} \\
&= \frac{\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{x})(Y_i - \bar{y})}{\frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{x})^2}
\end{aligned}
$$

2.2 Goodness of Fit & Correlation

Assume an underlying statistical model

$$y = \beta_0 + \beta_1 X + \epsilon$$

We use training data $(X_i, Y_i)_{i=1}^n$ to generate <u>prediction model</u> $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 X$
The formulas for $\hat{\beta}_0$ and $\hat{\beta}_1$ to minimize RSS:

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \begin{cases} \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{X} \\ \hat{\beta}_1 = \frac{\text{Cov}(X_i, Y_i)_{i=1}^n}{\text{Var}(X_i)_{i=1}^n} \end{cases}$$

<u>Goodness of fit</u>

$$\begin{aligned}
\text{RSS} &= \sum_{i=1}^n (y_i, \hat{y}_i)^2 \\
&= \sum_{i=1}^n (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i)^2 \\
&= \sum_{i=1}^n \Big((y_i - \bar{y}) - \hat{\beta}_1 (x_i - \bar{x}) \Big)^2 \\
&= \sum_{i=1}^n (y_i - \bar{y})^2 - 2\hat{\beta}_1 \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) + \hat{\beta}_1^2 \sum_{i=1}^n (x_i - \bar{x})^2 \\
&= \sum_{i=1}^n (y_i - \bar{y})^2 - \frac{\Big[\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y}) \Big]^2}{\sum_{i=1}^n (x_i - \bar{x})^2}
\end{aligned}$$

RSS scales with the number of data points n and also scales with y^2 units.
$\rightarrow$ We can fix the scaling and get out a goodness-of-fit score (R^2) between 0 and 1.

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}}$$

$$\text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2 \qquad \text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2$$

So for simple linear regression,

$$\begin{aligned} R^2 &= 1 - \frac{\text{RSS}}{\text{TSS}} \\ &= \frac{\text{TSS} - \text{RSS}}{\text{TSS}} \\ &= \frac{\left[\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})\right]^2}{\sum_{i=1}^n (x_i - \bar{x})^2 \sum_{i=1}^n (y_i - \bar{y})^2} \\ R &= \frac{\text{Cov}((x_i), (y_i))}{\text{Var}(x_i)^{1/2}\text{Var}(y_i)^{1/2}} \end{aligned}$$

Long story short R^2 is the fraction of variance explained by the model.

$\boxed{ex}$ $\quad \hat{\text{sales}} = 7.0 + 0.048 \cdot \text{TV}$
$R^2 = 0.61 = 61\%$ $\qquad r \approx 0.78$ pretty strong positive correlation

2.3 Regression Table

$\boxed{ex}$

	Coefficient	Std. error	t-statistic	p-value
Intercept (β_0)	7.0	0.46	15	<0.0001
TV (β_1)	0.048	0.0027	18	<0.0001

Std error is the error bar on linear coefficients.

$$\text{We know that } \begin{cases} \beta_0 \approx \hat{\beta}_0 \pm \text{SE}(\hat{\beta}_0) \\ \beta_1 \approx \hat{\beta}_1 \pm \text{SE}(\hat{\beta}_1) \end{cases}$$

Assume the underlying statistical model is true!

$$y_i = \beta_0 + \beta_1 X_i + \epsilon_i$$
$$\text{where } \epsilon_i \text{ are independent } \eta(0, \sigma^2) \text{ random variables}$$

If we generate N data points (x_i, y_i) from the model, then

$$\begin{cases} \text{Var}(\hat{\beta}_0)^{1/2} \approx \text{Std. } \hat{\text{error}}(\hat{\beta}_0) \\ \text{Var}(\hat{\beta}_1)^{1/2} \approx \text{Std. } \hat{\text{error}}(\hat{\beta}_1) \\ \mathbb{E}(\hat{\beta}_0) = \beta_0 \\ \mathbb{E}(\hat{\beta}_1) = \beta_1 \end{cases}$$